



The impact of an adult education voucher program: Evidence from a randomized field experiment[☆]

Guido Schwerdt^a, Dolores Messer^b, Ludger Woessmann^{c,*}, Stefan C. Wolter^d

^a Ifo Institute for Economic Research, CESifo, IZA, Poschingerstr. 5, 81679 Munich, Germany

^b University of Bern, Switzerland

^c University of Munich, Ifo Institute for Economic Research, CESifo, IZA, Poschingerstr. 5, 81679 Munich, Germany

^d Swiss Coordination Centre for Research in Education, University of Bern, CESifo, IZA; University of Bern, Centre for Research in Economics of Education, P.O. Box 8573, Schanzenekstrasse 1, 3001 Bern, Switzerland

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ABSTRACT

Lifelong learning is often promoted in aging societies, but little is known about its returns or governments' ability to advance it. This paper evaluates the effects of a large-scale randomized field experiment issuing vouchers for adult education in Switzerland. We find no significant average effects of the voucher program on earnings, employment, and subsequent education 1 year after treatment. But effects are heterogeneous: low-educated individuals are most likely to profit from adult education, but least likely to use the voucher. In addition, the public voucher program appears to crowd out firm-financed training. The findings cast doubt on the effectiveness of untargeted voucher programs in promoting labor market outcomes through adult education.

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1. Introduction

Governments in most countries support lifelong learning by intervening in the adult education market, using such mechanisms as tax subsidies, grants, loans, and subsidies to providers (OECD, 2004, 2005). Recently, several countries have introduced voucher programs

to stimulate adult education.¹ Vouchers are intended to increase the demand for adult education by reducing its direct cost while at the same time enhancing individuals' ability to exercise choice in the education market. However, little is known about the returns to adult education or the effectiveness of government attempts to promote adult learning. From a theoretical viewpoint, government intervention in the adult education market can be justified if market failures such as liquidity constraints, externalities, or imperfect information about private returns lead to suboptimal private investment in adult

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* Corresponding author.

E-mail addresses: schwerdt@ifo.de (G. Schwerdt), woessmann@ifo.de (L. Woessmann), stefan.wolter@vwi.unibe.ch (S.C. Wolter).

URL: <http://www.cesifo.de/woessmann> (L. Woessmann).

¹ In particular, education voucher models are provided by the regional chambers of trade in Austria, the Training Check in Belgium, adult education vouchers in several Italian regions, the North Rhine-Westphalian and Brandenburg Training Check in Germany, and the Training Check in the canton of Geneva in Switzerland. For example, since 2001 the Geneva canton provides the "Chèque annuel de formation", worth about 500 Euro on average, to all adults below a certain income threshold (of roughly 60,000 Euro). In recent years, more than 5000 vouchers were issued each year. In 2006, the German federal state of North Rhine-Westphalia introduced the "Bildungsscheck", worth up to 750 Euro, which could be used to pay for half of the cost of a training course for employees working in firms with fewer than 250 employees. In the first 1.5 years, around 140,000 vouchers, which are available to both employees and firms, were issued (Görlitz, 2010).

education (see McCall and Smith, 2009 for an overview). In such cases, public support for adult education should materialize in better labor market outcomes. On the other hand, government investment might be less effective in adult education than in early education (Cunha and Heckman, 2007). Any attempt to quantify the effects of adult education vouchers faces the fundamental problem of selection bias, because adults choosing to participate in education are likely to differ from adults choosing not to participate in many ways related to labor market success. Also, public programs may just crowd out existing training financed privately by individuals or firms.

This paper evaluates the labor market effects of a large-scale randomized field experiment that provided vouchers for fee-based adult education courses to a representative sample of the adult population in Switzerland. In the experiment, which we designed specifically for this purpose, a random sample of participants in the Swiss Labor Force Survey (SLFS), observed in 2005, was issued a voucher that could be redeemed for adult education courses in the first half of 2006. We exploit the exogenous variation generated by random assignment of the voucher to identify causal effects of the voucher treatment on earnings, employment status, and subsequent participation in adult education in 2007. In our estimation sample of more than 10,000 individuals observed from 2005 through 2007, the treatment group consists of 1422 individuals, which we can compare to a control group of 9099 individuals who did not receive the voucher offer. By making use of SLFS respondents and not releasing public information on the experiment, our setup ensured that neither treatment nor control group was aware of being part of an experiment, precluding Hawthorne effects.

The overall voucher effect is determined by the rate at which vouchers are picked up in the treatment group and by the effect of voucher-induced adult education on labor market success. The latter effect might differ from non-experimental estimates of the effects of adult education, which will be biased if selection into adult education is driven by non-ignorable unobserved characteristics. In addition, individuals who pick up the voucher may have non-average returns. We investigate this by comparing OLS estimates of the returns to adult education with IV estimates that use voucher receipt as an instrument for adult education. Moreover, we investigate potential heterogeneous effects of adult education and voucher receipt on labor market success for selected sub-groups of the population. We characterize the complier population in terms of these selected characteristics, which is valuable from a policy perspective as it describes the population actually affected by a voucher policy.

We find no statistically significant average effects of the voucher program on earnings and employment probabilities 1 year after treatment. The point estimates are, in fact, negative. While limited precision of IV estimates of the effects of voucher-induced adult education does not allow us to statistically reject the existence of small positive labor market effects, the IV point estimates imply substantially lower returns than suggested by OLS estimates. Analyzing the potential effect of the voucher on subsequent adult education, we can reject the possibility that the large persistence in adult education implied by cross-sectional OLS estimation is causal. Small positive effects of voucher-induced adult education on continuing human capital investments cannot be rejected and are in fact suggested by the IV point estimates.

The results highlight the importance of selection problems in estimating returns to public policies and suggest that the incidence of and returns to adult education vary between sub-populations. While our estimates identify a particular local average treatment effect (LATE) of voucher-induced adult education, this parameter has particular policy relevance because it captures what is achievable by a voucher policy. A characterization of the population of compliers based on the first-stage coefficients for specific sub-groups indicates that individuals with higher levels of education are more likely to comply with the voucher assignment. A sub-group analysis of labor

market effects suggests, however, that individuals with vocational training have a relatively high potential to benefit from participation in adult education. This substantial effect heterogeneity is a likely source for the statistical imprecision of our IV estimates of the average effect. Adult education vouchers might entail significant labor market effects when targeted at specific sub-groups.

Our findings cast doubt, however, on the effectiveness of untargeted voucher programs, as currently devised in many countries, to promote labor market outcomes through adult education. The same is likely to apply to the tax credit programs for education expenditures for adults as currently implemented in many countries, which are effectively untargeted as well.² Our results indicate that the public voucher program entails substantial deadweight losses because of voucher redemption by those who would have participated in adult education anyways, raising the overall cost of any public program. Furthermore, they suggest that the increased participation in self-financed adult education courses induced by the public voucher program comes partly at the detriment of adult training courses that would otherwise have been financed by employers. This result casts further doubt on the overall effectiveness of public policies in raising adult education and raises concerns that public voucher programs may partly crowd out more effective existing adult education modules.

While providing the first experimental evidence on labor market effects of an adult education voucher program aimed at the full adult population, our study adds to a large, albeit mostly non-experimental literature on labor market effects of adult education and training (see McCall and Smith, 2009 for a review). A first strand of this literature addresses the labor market returns to private on-the-job training within firms (e.g., Barron et al., 1997; Bassanini et al., 2005; Frazis and Loewenstein, 2005; Booth and Bryan, 2005), most of which is not very developed in terms of causal identification, though. For example, Leuven and Oosterbeek (2008) show that selection-correction and fixed-effects strategies usually employed in the literature are likely to remain subject to substantial selection bias. A second strand, again based on observational data, estimates labor market effects of formal education for adults, focusing on topics like the effects of upper-secondary or college attendance for returning adults (e.g., Leigh and Gill, 1997; Stenberg, 2011; Stenberg et al., 2011), often with a particular focus on displaced workers (e.g., Jacobson et al., 2005; Stenberg and Westerlund, 2008). A third strand focuses almost exclusively on the unemployed when estimating labor market impacts of public-sector job training as a means of active labor market policy (for surveys and meta-analyses, see Heckman et al., 1999; Card et al., 2010). While some studies evaluate randomized social experiments, the vast majority employs elaborate non-experimental identification techniques (e.g., Heinrich et al., 2009; Lechner and Wunsch, 2009). Still, the literature duly acknowledges that it is hard to ultimately rule out selection bias in such observational studies, and comparison to experimental evidence reveals that econometric evaluations of training programs can lead to very misleading results (see LaLonde, 1986).

Our study aims to extend the existing literature both methodologically and content-wise. Methodologically, we try to overcome remaining issues of selection into adult education by devising a randomized field experiment.³ Furthermore, the fact that job training

² In line with this, LaLumia (2012) does not find an effect of the United States Lifetime Learning tax credit and of tuition deductions on college attendance for most adults.

³ While there are few large-scale randomized experiments of public voucher programs in general, an example is the random lottery of housing vouchers in the Moving to Opportunity program (see Kling et al., 2007). In this regard, our study extends the experimental literature to public voucher programs subsidizing adult education. There is also an experimental literature on the effects of vouchers to attend private schools (e.g., Rouse, 1998; Peterson et al., 2003) based on lotteries of oversubscribed programs, but the extent to which evidence from oversubscribed samples generalizes to the total population – or from school children to adult population – is unclear. See Greenberg and Shroder (2010) for social experiments conducted in other areas and Harrison and List (2004) for an overview of field experiments in economics.

programs are highly targeted at the very poor or unemployed precludes conclusions about the relevance of targeting from existing comparisons of experimental versus non-experimental estimates like LaLonde (1986). By contrast, we provide such comparison for an untargeted intervention offered to a heterogeneous group of participants, where varying take-up by groups may matter. Such analysis thus indicates whether targeted programs might have higher potential.

In terms of content, there is a clear gap in the literature regarding public subsidization of self-financed education programs aimed at the adult population at large, in contrast to firm-financed adult education and to the population of unemployed. This is despite the fact that governments in several countries have implemented general voucher and tax credit programs for adult education. Our study provides first experimental evidence on the effect of publicly sponsored self-financed adult education for a sample representative of the entire population, thereby adding to the literature on designing public policies to promote adult education.

In what follows, the next section describes the large-scale randomized field experiment designed to offer vouchers for adult education. Section 3 reports descriptive evidence. Section 4 presents the estimation strategy and Section 5 the main results. Section 6 analyzes heterogeneity in effects and take-up of the voucher and Section 7 estimates whether the public voucher program crowds out firm-financed adult education. Section 8 concludes.

2. The experiment

The voucher experiment was initiated in 2005, when the Swiss government mandated a randomized field experiment with vouchers for adult education in order to investigate whether general voucher programs could increase adult education and improve labor market outcomes. The government was interested in whether a continuous updating of human capital could increase workers' productivity and reduce unemployment for those most at risk (see Wolter et al., 2003). In designing the experimental setup, we took advantage of the fact that the sample size of the Swiss Labor Force Survey (SLFS) had been raised significantly at the beginning of the century, but was scheduled to be reduced in 2006 because of financial constraints. The SLFS is structured as a rotating panel in which the respondents are interviewed for five consecutive years. As a result, about one-fifth of respondents are replaced every year. The financially induced reduction in the sample size in 2006 yielded the opportunity to select a random sample of 2437 individuals for the experiment who had been interviewed in the 2005 SLFS and would normally have been scheduled to continue participating in the next round of interviews but would have been dropped for financial reasons if our experiment had not funded their continuing participation. The randomization of our treatment group was performed separately by 40 SLFS regions.

The treatment group is matched by a control group of 17,234 individuals who had been interviewed by the SLFS in 2005 and were scheduled to be interviewed again in 2006. The experimental design enables us to use longitudinal data as well as cross-sectional information, among others on prior labor market outcomes and prior participation in adult education to check the experimental design and act as controls for pre-experimental status.

The only limitation that we imposed on the sample refers to age. Only individuals aged 20 to 60 were entitled to receive a voucher. Most under-20s would still be undergoing education or training, and a large share of over-60s would be likely to be retired pensioners. There were no limitations with respect to employment status, as increasing the skills of non-employed individuals may be a goal of the voucher policy.

The 2437 randomly selected individuals received a letter from the Swiss Federal Statistical Office during the first day of January 2006 containing the adult education voucher. The letter stated that the voucher was part of a project to promote lifelong learning by the

Federal Office for Professional Education and Training and that participants of the SLFS were particularly well suited to receive this gift. The letter was signed by the General Director of the Swiss Federal Statistical Office to eliminate any doubts about the legitimacy of the voucher. No public-domain information was generated at any point during the experimental period, to ensure that voucher recipients were unaware that the dispensing of the voucher was part of an observational study. There were vouchers of three different face values (worth 200, 750, and 1500 Swiss francs, respectively—one Swiss franc being roughly equivalent to one U.S. dollar), allocated randomly among the treatment group.⁴

Recipients were entitled to use the voucher for an adult education course of their choice which had to start in July 2006 at the latest. There were no restrictions on the content of the module because the intention was to use the choices as the basis for designing an efficient adult education voucher system later on. Subsequent inspection of course choices reveals that only 10.0% of those who picked up the voucher used it for course types that might broadly fall within the category of leisure courses. The largest category, at 39.7%, is courses that are directly job-related, followed by foreign language courses at 28.4% and IT courses at 21.0%. The vast majority of vouchers were thus used for courses that might be seen as labor market relevant. The average duration of the courses chosen by those who picked up the voucher was 42 h.⁵

Of the total of 2437 voucher recipients, 449 redeemed their vouchers, representing a redemption rate of 18.4%. To redeem the voucher, participants had to send the voucher with the course organizer's invoice to the Federal Office for Professional Education and Technology, which in turn paid out the amount to participants. The federal agency conducted spot checks to verify that the adult education uptake and invoices were bona-fide. Though redemption rates were lower at the beginning and end of the redemption period, the overall temporal pattern of redemptions does not suggest any unintentional special effects. Toward the end of June, the experimental subjects were surveyed for the SLFS as in previous years.⁶

Like other adult education statistics, the SLFS differentiates between different forms of formal and informal adult education, and participation rates vary accordingly. The crucial aspect of government funding is its intention to raise opportunities for attending fee-based courses. Therefore, the selected SLFS module was the one that asked participants subsequently whether they had attended fee-based courses within the past 12 months.

There are clear differences between, on the one hand, the adult education courses that the voucher program was aimed at and, on the other hand, formal post-compulsory education and public-sector training programs. Vouchers could not be used for tuition fees of programs that lead to degrees of the formal education system.⁷ In fact, formal education, including higher education, is virtually free of charge in Switzerland. While the voucher program targeted the whole adult population, programs like tax-based subsidies for formal education reach

⁴ In addition, half of the treatment group received an accompanying offer for free counseling, but we did not find any significant differences in effects between those who were offered counseling and those who were not.

⁵ We experimented with estimating effects for different course categories and durations, but results did not differ significantly between categories and durations. Similarly, qualitative results remained similar when dropping the leisure courses. Detailed results are available from the authors on request.

⁶ To enable later linkage of the survey data with the experimental data, participants were required at the end of the survey to consent to data linkage. This ensured both that the replies to the standard questions of the SLFS were not skewed because of linkage with the voucher, while at the same time addressing data protection concerns. Fortunately, none of the respondents objected to linkage of the SLFS data with the experiment.

⁷ Note that in Switzerland, as in most European systems of higher education, it is infeasible to participate in individual college courses and to accumulate them over many years in order to obtain a college degree. In fact, there are now strict time limits by which one must have finished the bachelor degree, which is usually within 1 or 2 years of the standard length of bachelor studies (of 3 years).

Table 1
Comparison of means for socio-demographic characteristics.

Variable (measured in 2005)	Original sample in 2005			Estimation sample in 2007		
	Treatment group (1)	Control group (2)	Difference (3)	Treatment group (4)	Control group (5)	Difference (6)
Adult education	0.437	0.428	0.009	0.454	0.435	0.020
Gross hourly earnings	38.432	38.657	−0.225	38.521	37.970	0.551
Employment	0.825	0.830	−0.005	0.825	0.824	0.001
Woman	0.557	0.542	0.015	0.565	0.559	0.007
Swiss	0.862	0.839	0.023***	0.900	0.865	0.035***
Age	41.426	41.644	−0.218	42.182	42.181	0.001
Compulsory education	0.146	0.154	−0.009	0.114	0.141	−0.027***
Vocational education	0.487	0.472	0.015	0.464	0.473	−0.010
Univ. entrance diploma	0.095	0.091	0.004	0.099	0.082	0.017**
Non-academic tertiary	0.156	0.151	0.005	0.195	0.162	0.033***
University	0.116	0.130	−0.014**	0.128	0.142	−0.014
Married	0.547	0.562	−0.015	0.606	0.608	−0.002
Divorced	0.128	0.119	0.009	0.142	0.139	0.003
Children of age <5	0.156	0.165	−0.009	0.146	0.156	−0.010
Household size	2.674	2.687	−0.013	2.788	2.738	0.051
Years of tenure	9.953	9.675	0.278	9.910	9.996	−0.086
Self-employed	0.169	0.163	0.006	0.164	0.161	0.003
Side job	0.067	0.069	−0.002	0.073	0.074	−0.001
Part-time	0.284	0.275	0.010	0.314	0.288	0.026**
Firm size <10	0.322	0.309	0.013	0.313	0.313	0.000
Firm size 10–49	0.272	0.274	−0.002	0.278	0.275	0.003
Firm size 50–99	0.104	0.110	−0.006	0.106	0.108	−0.003
Firm size ≥ 100	0.302	0.307	−0.005	0.303	0.304	−0.001
Number of observations	2437	17,234		1422	9099	

Note: Group means (each variable measured relative to its regional means). The first three columns refer to the original sample including all observations in 2005. The last three columns refer to the estimation sample including all observations in 2007 with non-missing information on adult education in 2006. All variables are measured in 2005. Data source: Swiss Labor Force Survey (SLFS) combined with data specifically collected in connection with the voucher experiment, which was conducted in 2006.

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

people with sufficient taxable income only, and only about a quarter of a cohort participates in tertiary education. The main difference to public-sector training programs is that the voucher program allowed individuals to enroll in any course of their choice, whereas public training programs usually recommend specific courses. Furthermore, the target group is very different—all adults in the voucher program vs. the unemployed in training programs.⁸ In principle, there may be overlap in course content between the voucher-financed adult education and public-sector training programs. In terms of scale, a voucher with a face value of 750 CHF would have been sufficient for an adult education program of roughly 20 lessons, equivalent to approximately half the length of an average attended course.

3. Descriptive evidence

This section provides descriptive statistics on several background characteristics and outcome variables. In 2007, the treatment and control groups were surveyed in the SLFS for the last time. All individuals of the original sample who were successfully interviewed in 2006 and 2007 constitute the estimation sample used in this analysis. Table 1 reports selected average characteristics separately for the treatment and control groups observed in 2005 and 2007.⁹ The

⁸ Public-sector training programs are very limited in extent in Switzerland, as the long-term unemployment rate is only around 1% of the employed workforce and the majority of long-term unemployed are older workers. Likewise, there are no higher education tax credits or deductions in Switzerland and almost no subsidized loans; only about 15% of Swiss higher-education students receive study grants. See OECD (2004, 2005) for the range of policies on adult education in other countries.

⁹ We use the terms treatment and control groups to refer to the experimental groups of voucher recipients and non-voucher recipients, although in the empirical analysis below the treatment effects we are interested in are in fact those of voucher-induced participation in self-financed adult education. The pattern of socio-demographic characteristics in 2006 (available from the authors on request) is broadly consistent with the 2005 pattern.

reported numbers of observations show substantial sample reduction relative to the full sample surveyed in 2005. While about one third of the total reduction in our estimation sample between 2005 and 2007 is due to standard rotation out of the SLFS, the remaining reduction reflects panel attrition of 17.8% on average over the 2 years. This extent of panel attrition is standard in the SLFS.¹⁰

The left panel of Table 1 shows that in most regards, the treatment group drawn randomly from the entire SLFS sample does not differ significantly from the control group.¹¹ In particular, there are no significant differences in the pre-treatment values of the outcome variables—adult education, earnings, and employment. There are small, but statistically significant differences only for 2 out of 20 variables, namely Swiss nationality and university education.¹² These variables were not considered by the SLFS in the randomization of our treatment group, so that the differences must be regarded as random events. What is important for the experiment is the fact that there

¹⁰ A model predicting attrition by the variables used as controls and outcomes in our analyses shows that characteristics such as female gender, Swiss nationality, age, family size, and higher education are positively associated with staying on in the panel. However, reassuringly these determinants of attrition do not differ significantly between pre-experimental and experimental-year waves. In addition, attrition does not depend significantly on our variables of labor market success, namely earnings and employment. Participation in adult education is positively associated with staying on in the panel, but again, this does not differ significantly between pre-experimental and experimental years. Detailed results are available from the authors upon request.

¹¹ Because of the regional randomization, variables are expressed relative to the mean of the respective region (equivalent to taking out region fixed effects).

¹² If the two categories of “non-academic tertiary” and “university” education, reported separately in the SLFS, are combined in a standard “tertiary education” category, the size of the combined category does in fact not differ significantly between treatment and control groups (in both panels of Table 1).

Table 2
Comparison of means for outcome variables.

Variable	Treatment group (1)	Control group (2)	Difference (3)
Adult education in 2006	0.436 (0.493)	0.330 (0.466)	0.106***
↪ At least one course self-financed (self-financed adult education in 2006)	0.318 (0.467)	0.181 (0.382)	0.137***
↪ At least one course firm-financed (firm training in 2006)	0.170 (0.377)	0.185 (0.386)	−0.015
Gross hourly earnings in 2007	39.729 (16.334)	39.455 (16.079)	0.273
Employment in 2007	0.837 (0.356)	0.833 (0.372)	0.004
Adult education in 2007	0.449 (0.493)	0.422 (0.490)	0.026*

Note: Group means (each variable measured relative to its regional means). Data source: Swiss Labor Force Survey (SLFS) combined with data specifically collected in connection with the voucher experiment, which was conducted in 2006. Sample: all observations in 2007 with non-missing information on adult education in 2006 (estimation sample). Earnings are measured in Swiss francs. Standard deviations in parentheses.

* $p < 0.10$.

** $p < 0.10$.

*** $p < 0.01$.

was no possibility of survey respondents to influence the assignment to the treatment or control group.¹³

Likewise, the pre-treatment values of the main outcome variables – adult education, earnings, and employment – do not differ significantly in our estimation sample of people observed from 2005 through 2007 (right panel of Table 1). The difference in the share of Swiss nationals remains. Due to slight differences in response patterns, the significance of differences in educational characteristics shifts to other categories in 2007. These observable differences caused by panel attrition and the initial over-assignment of vouchers to individuals with Swiss nationality emphasize the importance of controlling for these characteristics in the empirical analysis below.

Table 2 provides mean comparisons of outcome variables between treatment and control groups without controlling for any additional factors. The treatment group is 10.6 percentage points more likely to participate in fee-based adult education in the treatment year 2006.¹⁴ The inclusion of a specific adult education module in the 2006 wave of the SLFS allows us to distinguish between self-financed and firm-financed courses in 2006 (but not in other years). Voucher recipients are 13.7 percentage points more likely to participate in self-financed adult education in 2006, reflecting the first-stage relationship between voucher receipt and participation in self-financed adult education exploited in our IV approach. At the same time, in the unconditional descriptives the share of individuals who participated in firm-financed adult education in 2006 does not differ significantly between treatment and control groups.

¹³ When estimating our main specifications separately for the Swiss and immigrant sub-groups, the first-stage estimate is indistinguishable between the two sub-groups, indicating that immigrants respond to receiving a voucher in a similar way to Swiss nationals. The second-stage results for the Swiss sub-sample are very similar to the results for the combined sample reported below. While the second-stage results are also insignificant in the non-Swiss sub-sample, standard errors in this relatively small sample are too large to warrant detailed conclusions. Detailed results are available from the authors upon request.

¹⁴ There is a significant dip in reported participation rates in adult education in 2006 relative to previous and successive years that was a one-year event and concerned all groups of survey participants in a similar way. According to an official statement of the Federal Office for Statistics (<http://www.bfs.admin.ch/bfs/portal/de/index/themen/15/07/ind19.informations.190104.html>), the dip is likely to reflect a change in the reporting behavior of survey respondents caused by a change in the sequencing and wording of questions due to the inclusion of the specific adult education module in the 2006 wave of the SLFS, rather than any real dip in overall participation in adult education. As these changes in the 2006 questionnaire affected the treatment and control groups in the same way, it does not raise concerns for our identification strategy.

The bottom part of Table 2 reports group means for our three outcome variables in 2007: gross hourly earnings, employment status, and participation in adult education. Gross hourly earnings are computed by dividing gross monthly earnings by effective hours worked. Employment status is a dummy variable indicating whether an individual was employed at the time of the survey. There are no significant mean differences in gross hourly earnings and employment probabilities in 2007 between treatment and control groups. Fig. 1 also provides kernel-density estimates of the distribution of gross hourly earnings for treatment and control groups measured in 2005, 2006, and 2007. As the reported p -values from two-sample Kolmogorov–Smirnov tests of equality of distributions indicate, we can reject the hypothesis that the wage distributions are different at any point.

Table 2 reveals, however, a marginally significant difference in participation in adult education in 2007. Those who received a voucher in 2006 are 2.6 percentage points more likely to participate in adult education in 2007, the year after the voucher redemption period. Whether this indicates a positive effect of voucher-induced participation in 2006 on subsequent private investments in adult education or is a corollary of the slight difference in pre-treatment participation in the estimation sample (right panel of Table 1) is the subject of the more rigorous analysis of the effects of voucher-induced adult education below.

4. Empirical strategy

From a public policy perspective, our main interest is in the effect of the adult education voucher program on subsequent labor market outcomes. Because of the random assignment of vouchers in our field experiment, we can obtain an unbiased estimate of the effect of providing adult education vouchers by regressing indicators of labor market success on a dummy indicating voucher receipt:

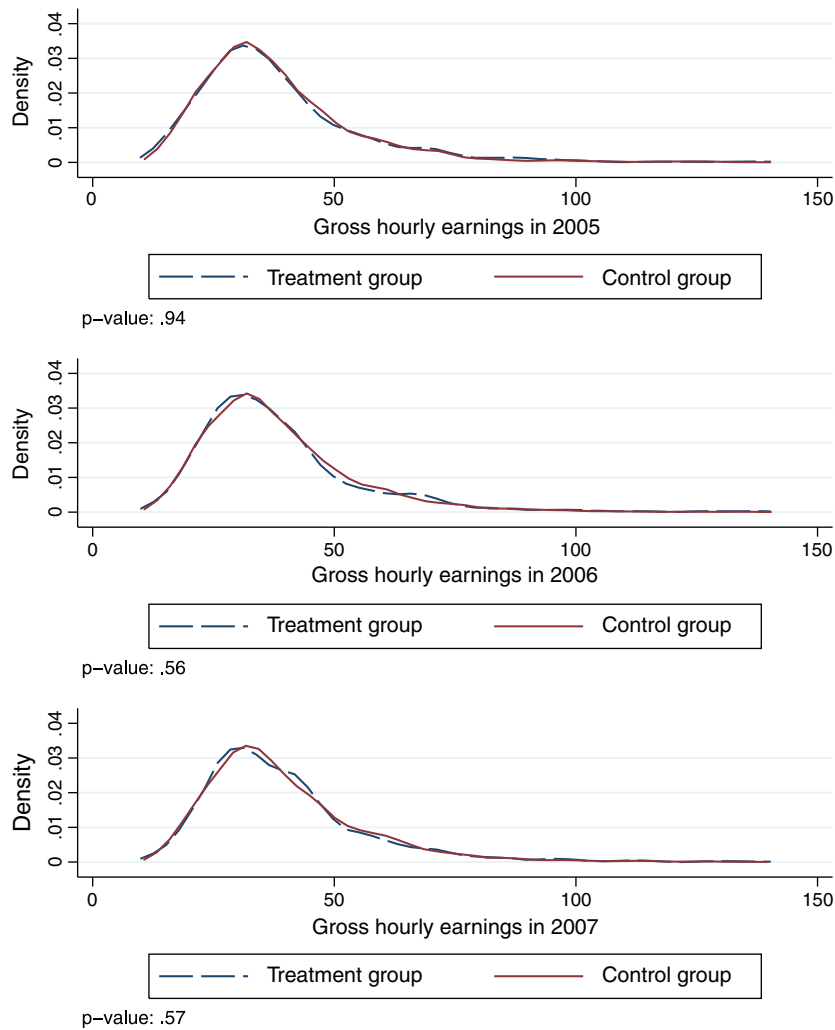
$$y_{i,t+1} = \phi_i \text{voucher}_{i,t} + \lambda_0' Z_i + \varepsilon_{i,t}, \quad (1)$$

where $y_{i,t+1}$ is the outcome of interest for individual i at time $t+1$, $\text{voucher}_{i,t}$ is an indicator of whether individual i received a voucher at time t , Z_i is a vector of control variables, and $\varepsilon_{i,t}$ is an error term uncorrelated with all right-hand side variables.

The parameter of interest is ϕ_i . It measures the effect of receiving an adult education voucher on the outcome of interest. While in most models, we will assume that $\phi_i = \phi$ for all i , in some analyses we will allow the effect to be heterogeneous across voucher values and across sub-groups of individuals based on their education and gender. Because not every voucher recipient uses the voucher, ϕ_i identifies an intention-to-treat (ITT) effect of the voucher policy. While in principle, the experimental setup would not require considering control variables, the limitations in the initial randomization and subsequent attrition discussed in the previous section lead us to control for covariates in our models. To account for as many factors as possible, the vector of control variables does not only include background characteristics measured at t , but also pre-treatment values of background and outcome variables at $t-1$. Note, however, that qualitative results are unchanged when estimating the models without any covariates.¹⁵

The overall effect of the voucher program is a combination of the rate at which recipients use the voucher and of the effect of participation in an adult education course on future labor market success for these voucher users. As seen above, although voucher receipt did positively affect immediate participation probabilities in adult education courses, pick-up rates are quite small. Whether the adult education voucher program positively affects labor market outcomes thus crucially depends on the returns to adult education for those individuals

¹⁵ Detailed results are available from the authors on request.



Note: Kernel density estimates based on 2007 estimation sample. p-values refer to two-sample Kolmogorov-Smirnov tests of the equality of distributions. Datasource: Swiss Labor Force Survey (SLFS) combined with data specifically collected in connection with the voucher experiment, which was conducted in 2006.

Fig. 1. Wage distributions by year and voucher status.

who complied with the voucher assignment and used the offered voucher.

We therefore perform more detailed analyses of the effect of adult education on future labor market success. In the conventional literature on labor-market returns to adult education, the following cross-sectional regression model is used to link labor market success to participation in adult education and a set of exogenous covariates:

$$y_{i,t+1} = \beta_i \text{adult_education}_{i,t} + \alpha'_0 X_{i,t} + \eta_{i,t}, \quad (2)$$

where $\text{adult_education}_{i,t}$ is an indicator of (self-financed) adult education in t and $X_{i,t}$ is a vector of background characteristics measured at t . Since we are interested in the effects of employee-financed adult education and there is also firm-financed adult education and training, the control vector $X_{i,t}$ includes an indicator for the latter.¹⁶ The parameter of interest, β_i , measures the association between participation in self-financed adult education courses and labor market success

¹⁶ Our main results on self-financed adult education are qualitatively the same in specifications that do not control for firm training and in specifications that define the firm-training variable as having participated only in firm-financed education courses.

in the subsequent period. Again, we will present models that assume $\beta_i = \beta$ for all i , as well as models with effect heterogeneity by education and gender. In standard cross-sectional work, $\hat{\beta}^{OLS}$ is likely to be a biased estimate of the effect of interest because participation in adult education is selective, partly in response to heterogeneity in β .

Longitudinal data can help to address part of the selectivity bias. We can exploit the panel structure of our data and include past outcomes in Eq. (2). The simplest way to do so would be to rewrite Eq. (2) in terms of a value-added model with first-differences of the outcome as dependent variable. An alternative and less restrictive specification is to include past outcomes as controls. In this case, the equation of interest becomes:

$$y_{i,t+1} = \beta_i \text{adult_education}_{i,t} + \alpha'_0 X_{i,t} + \alpha'_1 X_{i,t-1} + \alpha_2 y_{i,t-1} + \eta_{i,t}, \quad (3)$$

where $X_{i,t-1}$ is a vector of controls measured at $t-1$ and $y_{i,t-1}$ is the outcome measured at $t-1$. The inclusion of past outcomes should eliminate the confounding effect of time-invariant individual characteristics that affect labor market outcomes. Yet, selection into adult education might not only be based on levels of observed or unobserved variables. For instance, if selection into self-financed adult

education is driven by trends in the outcome variable, the estimate for β_i in Eq. (3) provides a biased estimate of the effect of interest.

The exogenous variation in adult education generated by the random assignment of the voucher in our experiment can be exploited to solve the endogeneity of self-financed adult education. As vouchers are assigned randomly, we can estimate a two-stage least-squares (2SLS) model with voucher receipt as instrument. In the just-identified model where $voucher_{i,t}$ is the only instrument for the endogenous variable $adult_education_{i,t}$, the first-stage model relating voucher receipt to participation in adult education is:

$$adult_education_{i,t} = \gamma_i voucher_{i,t} + \pi'_0 X_{i,t} + \pi'_1 X_{i,t-1} + \pi'_2 y_{i,t-1} + \mu_{i,t}, \quad (4)$$

where γ_i is the first-stage effect of the instrument. The parameter corresponds to the pick-up rate of the voucher. The resulting IV estimate, $\hat{\beta}^{IV}$, is an unbiased estimate for the average effect of self-financed adult education on labor market success for compliers. It identifies the local average treatment effect (LATE) for individuals who participate in adult education when receiving a voucher, but would not have participated if they had not received the voucher (Imbens and Angrist, 1994). Note that the effect of voucher-induced adult education on labor market success, β^{IV} , corresponds to the ITT effect of the voucher policy, ϕ (which reflects the reduced form of the IV model), divided by the pick-up rate estimated by the first-stage coefficient, γ .

5. Main results

The discussion of empirical results starts with the effects of the voucher program on participation in adult education and on labor market outcomes. Given that part of the overall program effect is driven by the pick-up rate of the voucher, the second part of the section further investigates what our experimental setting reveals about the labor market effects of adult education. We investigate the effects defined in Section 4 based on three labor market outcomes: hourly earnings, employment, and subsequent participation in adult education. All outcomes are measured in 2007. Given that the voucher treatment took place in the first half of 2006, the results should be interpreted as estimates of short-run effects.

5.1. The impact of the adult education voucher program

5.1.1. Effects of the voucher program on participation in adult education

We start by investigating the extent to which the voucher program affected participation in adult education courses in 2006, exploiting the random assignment in our experimental study to identify exogenous variation in voucher receipt. That is, we estimate Eq. (4) which reveals the rate at which vouchers have been picked up. In the first column of Table 3, the estimated coefficient on the dummy variable indicating voucher receipt identifies the causal effect of the voucher program on participation in adult education in 2006. The coefficient of 0.129 corresponds to the pick-up rate of the voucher. The estimate shows that receiving a voucher increases the probability to participate in an adult education course in 2006 by 13 percentage points. This effect is highly significant. That is, the voucher program did have a significant impact on participation in adult education.

The marginal effect of the voucher program on participation in adult education (12.9 percentage points) is substantially lower than the voucher redemption rate (18.4%). This shows that there is a relatively large share of always-takers who would have participated in adult education courses even without the voucher program. This raises the cost of the public voucher program beyond the direct outlays for the additional voucher-induced participation, as it entails a deadweight loss of about 30%.

Table 3

The impact of the adult education voucher program.

	Self-financed adult education in 2006 (1)	Earnings in 2007 (2)	Employed in 2007 (3)	Adult education in 2007 (4)
Voucher in 2006	.129*** (.011)	-.0006 (.0073)	-.0005 (.0080)	.013 (.013)
Firm training in 2006	-.011 (.010)	.030*** (.006)	.029*** (.007)	.233*** (.012)
Employed in 2005	.023** (.011)		.592*** (.008)	
Earnings in 2005		.537*** (.011)		
Adult education in 2005				.270*** (.009)
Woman	.117*** (.008)	-.041*** (.007)	-.035*** (.006)	.047*** (.009)
Swiss	.044*** (.013)	.023*** (.008)	-.005 (.009)	.048*** (.014)
Age	.005* (.003)	.017*** (.002)	.016*** (.002)	.016*** (.004)
Age ² /10 ³	-.077** (.037)	-.173*** (.026)	-.221*** (.026)	-.209*** (.042)
Region	Yes	Yes	Yes	Yes
Language	Yes	Yes	Yes	Yes
Family	Yes	Yes	Yes	Yes
Education	Yes	Yes	Yes	Yes
Job char. 2005	No	Yes	No	No
Observations	10,521	5606	10,521	10,515
R-squared	.072	.610	.446	.205

Note: Dependent variable indicated in the first row (log gross hourly earnings in column 2). Standard errors in parentheses. Regional controls include 40 dummies for region of residence. Family controls include household size and indicators for being married, being divorced, and the number of children under 5. Education controls include indicators for non-academic tertiary degree, vocational education, university entrance diploma, compulsory education, and university degree. Job characteristics in 2005 include tenure as well as dummy variables for self-employment, part-time work, having a side job, 3 firm size categories, and industry based on 12 NOGA categories. Data source: Swiss Labor Force Survey (SLFS) combined with data specifically collected in connection with the voucher experiment, which was conducted in 2006.

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

In the voucher program, vouchers with three different face values were randomly rotated within the treatment group: 200, 750, and 1500 Swiss francs, respectively. Table 4 reports estimates separately for the three face values. The pattern of point estimates of the effect of voucher receipt on participation in adult education in 2006 in the first column is consistent with a significantly lower pick-up rate for vouchers with the lowest face value, but no significant difference between the two higher face values.

5.1.2. Effects of the voucher program on earnings

Earnings are the most obvious measure of labor market success. Therefore, column 2 of Table 3 reports the estimate of the causal effect of voucher receipt on log gross hourly earnings in 2007 based on Eq. (1). This intention-to-treat (ITT) effect of the voucher policy is well identified because of the random assignment of the voucher.

The point estimate of the effect of voucher receipt on earnings is in fact negative, close to zero (at -0.0006), and statistically insignificant. With the given standard errors, we can reject that the voucher increased earnings on average by more than 1.14% with 90% confidence (1.37% at 95% confidence). What does this level of statistical power mean? Consider a politician who expects that by handing out free adult education, she gets universal take-up and an overall wage increase for her population of 1/8 of a standard rate of return to a year of education of about 10% (e.g., Card, 1999), because the average course duration (of 42 h) is roughly equivalent to 1/8 of a full year of higher education. This 1.25% increase is exactly the order of magnitude that can be ruled out with high confidence based on our

Table 4
The impact of the voucher program by face value of the voucher.

	Self-financed adult education in 2006 (1)	Earnings in 2007 (2)	Employed in 2007 (3)	Adult education in 2007 (4)
[a] 200 Swiss francs face value	.098*** (.019)	-.011 (.013)	-.010 (.014)	-.011 (.022)
[b] 750 Swiss francs face value	.144*** (.018)	-.0002 (.0117)	-.003 (.013)	.028 (.021)
[c] 1500 Swiss francs face value	.140*** (.018)	.008 (.012)	.010 (.013)	.021 (.021)
Prob > F				
Test: [a] = [b]	.067	.537	.710	.182
Test: [a] = [c]	.095	.284	.275	.272
Test: [b] = [c]	.865	.633	.463	.807
Test: [a] = [b] = [c]	.131	.562	.537	.367

Note: ITT estimates of having received a voucher in 2006 for subgroups of the voucher recipients defined by the face value of the voucher reported in the first column. The first row indicates the dependent variable. The last four rows indicate p-values of F-tests of the hypothesis that different voucher values have identical treatment effect. Standard errors in parentheses. Specifications correspond to the specifications presented in Table 3. Data source: Swiss Labor Force Survey (SLFS) combined with data specifically collected in connection with the voucher experiment, which was conducted in 2006.

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

randomized evaluation. Note again, though, that the point estimate (rather than the upper bound of the confidence interval) of the effect of the voucher program on earnings is in fact slightly below zero.

All our models control for firm training in 2006. In Section 7 below, we show that voucher-financed adult education in fact partly crowds out firm-financed programs. This part of the variation in firm training is endogenous to the voucher program, so that the presented models show the voucher effect after shutting down the channel of reduced firm training. Given that firm training itself is positively related to labor market outcomes, the presented models provide conservative estimates of the overall labor market effects of the voucher program. Thus, models that do not control for firm training (not shown) throughout yield even more negative assessments of the voucher effects; however, the estimates are very close to the ones presented here, and qualitative results are totally unaffected.¹⁷ For example, the point estimate of the effect of voucher receipt on earnings in a model without firm training control is -0.0011 (standard error 0.0074).

The pattern of point estimates for the three face values in Table 4 points to a weak (albeit insignificant) positive association of face values with labor market effects. Still, results are statistically insignificant and close to zero even for the vouchers with the highest face value.

5.1.3. Effects of the voucher program on employment

In a rigid labor market, effects of adult education might materialize more in terms of higher employment probabilities than in terms of increased earnings. Moreover, the earnings results are limited to continuously employed individuals. Participation in adult education might be particularly beneficial for the non-employed (Stenberg and Westerlund, 2008).

Column 3 of Table 3 reports the estimated effect of voucher receipt on an indicator for employment in 2007. Again, the point estimate is slightly negative (-0.0005) and not statistically significantly different from zero. The effect is, however, estimated with relatively low precision. The upper 90-percent confidence bound lies at 0.0127, which implies that we can confidently reject only positive effects

exceeding a 1.27 percentage-point increase in future employment probabilities.

5.1.4. Effects of the voucher program on subsequent adult education

While it seems reasonable to expect that participation in adult education courses should have its most imminent effect in the year after participation, long-term effects might exceed short-run returns if participation in adult education is continuous rather than nonrecurring. For example, if imperfect information about returns to adult education leads to suboptimal private investment and individuals learn about the true returns by participating once, voucher-induced participation might induce further subsequent investments. While we do not have information on labor market outcomes beyond 2007, we can analyze effects of the voucher program on participation in adult education in 2007, when the voucher was no longer valid.

Column 4 of Table 3 reports the corresponding reduced-form estimate of receiving a voucher in 2006 on participation in adult education in 2007. The effect is statistically insignificant with a point estimate of 0.013. The upper bound of the 90 (95) percent confidence interval is at an effect size of 3.4 (3.8) percentage points additional participation in adult education after the voucher period. A careful interpretation of this result concludes that, while we cannot reject that there is no effect of the voucher program on adult education beyond the voucher period, we also cannot reject small positive effects of the voucher on subsequent private investments in adult education.¹⁸

Summing up, the results suggest a pessimistic view on the effectiveness of adult education voucher programs. The estimates foster the concern that the impact of the voucher policy on success in the labor market is close to zero in the short run. The voucher program might entail small positive effects on subsequent participation in adult education, but these are very unlikely to affect more than 4% of the population.

5.2. Evidence on the returns to adult education

The overall effect of the voucher program combines the pick-up rate with the labor market return to adult education. Given the low rate of 13 percentage points at which the voucher induced additional participation in adult education (Section 5.1.1), the success of any voucher policy crucially depends on the size of the labor market returns. Thus, we next provide further detail on what our experimental data reveal about the labor market returns to adult education.

5.2.1. Adult education and earnings

Inspired by the seminal work of Mincer (1974), a vast literature on the estimation of causal effects of education on earnings emerged (Card, 1999). Yet, most studies focus on returns to schooling, with thin evidence on the returns to adult education. In particular, little evidence exists on the causal effect of self-financed adult education. We thus start our analysis of labor market returns with a replication of standard non-experimental specifications.

Table 5 displays results with log gross hourly earnings in 2007 as dependent variable. The first four columns show results of OLS estimations based on Eq. (2) with different sets of covariates. In the first column, earnings are regressed on a dummy indicating participation in self-financed adult education in 2006, a dummy indicating participation in firm-financed adult education in 2006, a gender dummy, a dummy for Swiss nationality, a linear and quadratic term in age, and region indicators. The results indicate positive associations of earnings with both self-financed and firm-financed adult education. The association between earnings and adult education is, however, stronger in the case of firm-financed education. This is in line with previous results in the literature suggesting significant positive

¹⁷ This is true for all models in Tables 3–9; detailed results available from the authors on request.

¹⁸ In fact, in a model without any control variables, the estimate on voucher receipt is marginally significant.

Table 5
The effect of adult education on earnings.

	OLS				IV	
	(1)	(2)	(3)	(4)	1st stage (5)	2nd stage (6)
Self-financed adult education in 2006	.040*** (.009)	.038*** (.009)	.005 (.009)	.008 (.007)		–.005 (.061)
Voucher in 2006					.119*** (.015)	
Firm training in 2006	.112*** (.009)	.108*** (.008)	.085*** (.008)	.030*** (.006)	–.003 (.012)	.030*** (.006)
Earnings in 2005				.537*** (.011)	.040* (.021)	.538*** (.011)
Woman	–.164*** (.007)	–.163*** (.007)	–.128*** (.007)	–.042*** (.007)	.078*** (.013)	–.041*** (.008)
Swiss	.111*** (.011)	.082*** (.011)	.056*** (.011)	.023*** (.008)	.028* (.017)	.023*** (.009)
Age	.046*** (.003)	.047*** (.003)	.040*** (.003)	.017*** (.002)	–.004 (.004)	.017*** (.002)
Age ² /10 ³	–.471*** (.034)	–.475*** (.035)	–.391*** (.032)	–.175*** (.026)	.017 (.051)	–.175*** (.026)
Region	Yes	Yes	Yes	Yes	Yes	Yes
Language	No	Yes	Yes	Yes	Yes	Yes
Family	No	Yes	Yes	Yes	Yes	Yes
Education	No	No	Yes	Yes	Yes	Yes
Job char. 2005	No	No	No	Yes	Yes	Yes
Observations	5606	5606	5606	5606	5606	5606
R-squared	.209	.224	.355	.610	.095	.610
MES	–	–	–	–	65.99	–

Note: Dependent variable: log gross hourly earnings in 2007 (in column 5: self-financed adult education in 2006). Standard errors in parentheses. Regional controls include 40 dummies for region of residence. Family controls include household size and indicators for being married, being divorced, and the number of children under 5. Education controls include indicators for non-academic tertiary degree, vocational education, university entrance diploma, compulsory education, and university degree. Job characteristics in 2005 include tenure as well as dummy variables for self-employment, part-time work, having a side job, 3 firm size categories, and industry based on 12 NOGA categories. MES refers to the Cragg and Donald (1993) minimum eigenvalue statistic. Data source: Swiss Labor Force Survey (SLFS) combined with data specifically collected in connection with the voucher experiment, which was conducted in 2006.

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

effects of work-related training on earnings (e.g., Bassanini et al., 2005; Frazis and Loewenstein, 2005). Descriptively, participation in self-financed adult education is associated with 4.0% higher earnings and firm-financed education with 11.2% higher earnings in the following year. Both adult education coefficients decrease slightly when we additionally control for family characteristics and differences in language of the interview (column 2).

However, when adding controls for educational attainment in column 3 of Table 5, the coefficient on self-financed adult education drops to close to zero and loses statistical significance. The coefficient on self-financed education remains close to zero in models based on Eq. (3) which additionally controls for log gross hourly earnings in 2005 and other job characteristics measured in 2005, before the adult education measure (column 4). In this specification, though, the coefficient on firm-financed training is reduced by more than half. Overall, the results reveal that the positive association between adult education and earnings typically found in the cross-sectional literature is likely to mostly reflect selective participation in adult education based on characteristics observed in our study.¹⁹

The OLS coefficient provides an unbiased estimate of the causal effect only if the selection-on-observables assumption holds, that is,

if participation in self-financed adult education is uncorrelated with unobserved determinants of earnings. On the one hand, this seems reasonable as the specification estimated in column 4 of Table 5 includes a large set of covariates relevant for the decision to participate in adult education—much richer than commonly available in the literature. In particular, the specification includes past outcomes, thereby controlling for heterogeneity among individuals typically unobserved in the cross-section. Moreover, our data allow us to distinguish self-financed from firm-financed adult education. The results in column 4 show that this distinction matters, with the coefficient on employer-financed adult education remaining significant at 3% whereas the coefficient on adult education paid by the individual is insignificant and close to zero. This is likely to reflect differences in the selection into these two types of adult education. On the other hand, selection into adult education might still be influenced by other unobserved factors. In particular, participation in self-financed adult education might be partly driven by expected earnings growth (Pischke, 2001). In this case, the estimate would still be biased.

In our experimental study, the random assignment of vouchers offers a source of exogenous variation that we can exploit to identify the causal effect of self-financed adult education. Voucher receipt is a valid instrument for self-financed adult education: It satisfies the exogeneity assumption by construction and is also relevant. Again, to account for limitations in the initial randomization and subsequent attrition (see Section 3), all IV models include the set of covariates shown in Table 5, but substantive results of the IV estimations are the same in models without any control variables.²⁰

¹⁹ To make the OLS estimates conditional on both earnings level and earnings growth, we can also add earnings growth between 2004 and 2005 as a control. This does not significantly alter the main findings (not shown). As long as 2005 earnings are not included in the model, earnings growth between 2004 and 2005 does not enter the model significantly. But once the positive association between 2005 earnings and 2006 earnings is controlled for, 2004-to-2005 earnings growth is significantly negatively associated with 2006 earnings, which may reflect that a pay rise in one period makes another one in the subsequent period less likely.

²⁰ Detailed results are available from the authors on request.

The estimated first-stage model of the IV approach specified in Eq. (4) is reported in column 5 of Table 5.²¹ The result confirms that the significant effect of voucher receipt on participation in adult education shown in the first column of Table 3 holds in the sample of employed persons. Using weak instruments not only produces less precise estimates, but can also lead to large inconsistencies of the IV estimates (Bound et al., 1995). With only one endogenous regressor, the Cragg and Donald (1993) minimum eigenvalue statistic reported in the last row of column 5 is identical to the *F*-statistic on excluded instruments, which Stock et al. (2002) suggest must be large, typically exceeding 10, for 2SLS inference to be reliable. A value of 61.5 clearly exceeds this threshold. But Monte Carlo simulations show that simply looking at the *F*-statistic might not be sufficient (Hall et al., 1996). Stock and Yogo (2005) suggest a more formal test for weak instruments where critical values for worst-case rejection rates of 5, 10, 20, and 25% for nominal 5% Wald tests of the endogenous regressors are provided. In our case, the minimum eigenvalue statistic exceeds the critical value at all levels of significance. We can, thus, conclude that voucher receipt is not a weak instrument.

Column 6 of Table 5 shows the corresponding second-stage IV estimate of the effect of self-financed adult education on future earnings. The point estimate of -0.005 is small, negative, and statistically insignificant. This estimate is well below the estimate of the initial OLS specifications. However, the standard error of the IV estimate is large, indicating that our IV approach lacks precision. In particular, we cannot statistically reject the existence of effect sizes in the magnitude of the initial OLS coefficients. In fact, the standard error of 6% is about five times the effect size of adult education (of 1.25%, see above) that might be expected based on estimated returns to schooling. The IV model would thus be very unlikely to detect any usual effect of adult education. The lack of precision of the IV estimate results from the relatively high share of always-takers, and possibly from heterogeneity in the effect. However, the substantial reduction in the point estimate relative to the OLS results might still be economically important. The reduction in the estimated return to adult education may have two reasons: first, standard OLS results might be upward biased because individuals with favorable characteristics predominantly select themselves into self-financed adult education. The fact that OLS estimates with extensive controls including the lagged dependent variable are already close to zero corroborates this interpretation. Second, considering the LATE interpretation of our IV approach, the IV estimate might measure the effect for a particular group of individuals with low returns to adult education in terms of future earnings.

An additional reason for the difference between the sizeable positive estimate in the standard OLS models and the zero point estimates in the IV specification and in the OLS model with lagged dependent variable might be that the former partly captures a multi-year treatment, because participation in adult education is correlated over time. Correlation coefficients of adult education across years are between 0.3 and 0.4 for the 2004–2007 waves. Thus, when adding adult education in 2005 as an additional control to the specifications in Table 5, it enters significantly and renders the estimate on self-financed adult education in 2006 smaller and statistically insignificant already in the specifications of the first two columns (not shown). Similarly, when further adding adult education in 2004 to the models, it tends to capture additional significance (over and above adult education in 2005) in the basic OLS models, although no longer when earnings in 2005 are controlled for. Unfortunately, because the data do not allow us to differentiate between self-financed and firm-financed education in years other than 2006, the

dominance of the 2005 and 2004 education variables might also capture a dominance of firm-financed training, though.

5.2.2. Adult education and employment

OLS and IV estimates of the effect of adult education on employment are reported in Table 6.²² The structure of Table 6 resembles the structure of Table 5 and follows the empirical strategy laid out in Section 4. As, in contrast to the earnings equations of Table 5, here we do not condition on continuous employment, all estimations are based on the entire estimation sample. Consequently, we also do not include job characteristics in 2005 as covariates.

In the first three columns of Table 6, we estimate employment regressions based on Eq. (2) with different sets of controls. In column 4, we additionally control for employment status in 2005. The OLS estimates reveal a similar pattern as in the case of earnings: self-financed adult education is positively associated with future employment probabilities, which in the case of employment remains significant even in the most extensive OLS model. However, selection into adult education matters: as we extend the control model, the estimated associations of employment with both self-financed and firm-financed adult education decrease. Again, the effects for firm-financed adult education appear to be larger. Results displayed in column 4 indicate that adult education is related to 1.5 percentage-points higher employment probabilities in 2007 in the case of self finance and to 2.9 percentage-points higher employment probabilities in the case of firm finance.²³

Columns 5 and 6 of Table 6 present results of estimating an IV approach with voucher receipt as instrument using 2SLS.²⁴ The first-stage estimation replicates the result already shown in Table 3. The IV estimate of the effect of self-financed adult education on employment probabilities in 2007 is small, negative, and statistically insignificant. Standard errors are, however, again large and do not allow to reject the OLS estimates. Similar to the interpretation of the earnings results, the decrease in the point estimate relative to OLS still provides some indication that individuals complying with the voucher assignment might not experience any effect of adult education on future employment probabilities.

5.2.3. Subsequent adult education

The OLS estimates of the effect of self-financed adult education in 2006 on subsequent adult education in 2007 in Table 7 show that participation in adult education is indeed highly persistent. Estimates in column 1 show that 31% of the individuals who participated in self-financed adult education in 2006, and 34% of those who participated in firm-financed adult education in 2006, continue to participate in adult education in 2007 (after controlling for gender, age, and nationality). Adding further covariates to the model reduces the coefficients slightly, to 0.28 in the case of self-financed adult education (column 3).

Part of the observed correlation between adult education in 2006 and in 2007 might simply reflect habit formation or might be caused by adult education modules lasting for more than 1 year. To control for such confounding factors, we add participation in adult education in 2005 to the model. The results presented in column 4 of Table 7

²² All estimations are based on linear models despite the binary nature of the outcome variable. We regard the OLS results as a linear approximation of the conditional expectation function of interest, as suggested in Angrist and Pischke (2009). Probit estimations produce almost identical results for the effects of interest.

²³ Since one must be employed to receive firm-financed training, controlling for firm-financed training in 2006 might indirectly capture effects of employment in 2006. However, results of Table 6 models without controlling for firm-financed training in 2006 are very similar (not shown).

²⁴ Nonlinear second-stage estimates with continuous or multi-valued regressors require a correctly specified functional form in order to interpret the estimates easily. We, therefore, apply a 2SLS procedure although the dependent variable is binary. Even if the underlying second-stage relationship is nonlinear, 2SLS typically captures an average effect of economic interest analogous to the LATE parameter for dummy endogenous regressors (Angrist and Imbens, 1995).

²¹ We model the first stage as linear despite the binary nature of the outcome variable. Using a linear regression for the first-stage estimates generates consistent second-stage estimates even with a dummy endogenous variable (Angrist and Krueger, 2001).

Table 6
The effect of adult education on employment.

	OLS				IV	
	(1)	(2)	(3)	(4)	1st stage (5)	2nd stage (6)
Self-financed adult education in 2006	.044*** (.009)	.038*** (.009)	.026*** (.009)	.015** (.007)		–.004 (.062)
Voucher in 2006					.129*** (.011)	
Firm training in 2006	.135*** (.009)	.128*** (.009)	.116*** (.009)	.029*** (.007)	–.011 (.010)	.029*** (.007)
Employed in 2005				.592*** (.008)	.023** (.011)	.592*** (.008)
Woman	–.139*** (.007)	–.139*** (.007)	–.122*** (.007)	–.036*** (.006)	.117*** (.008)	–.034*** (.009)
Swiss	.044*** (.010)	.013 (.011)	–.001 (.011)	–.006 (.009)	.044*** (.013)	–.005 (.009)
Age	.032*** (.003)	.039*** (.003)	.036*** (.003)	.016*** (.002)	.005* (.003)	.016*** (.002)
Age ² /10 ³	–.410*** (.032)	–.495*** (.033)	–.462*** (.033)	–.220*** (.026)	–.077** (.037)	–.221*** (.027)
Region	Yes	Yes	Yes	Yes	Yes	Yes
Language	No	Yes	Yes	Yes	Yes	Yes
Family	No	Yes	Yes	Yes	Yes	Yes
Education	No	No	Yes	Yes	Yes	Yes
Observations	10,521	10,521	10,521	10,521	10,521	10,521
R-squared	.101	.117	.128	.446	.072	.446
MES	–	–	–	–	133.78	–

Note: Dependent variable: employment dummy in 2007 (in column 5: self-financed adult education in 2006). Standard errors in parentheses. Regional controls include 40 dummies for region of residence. Family controls include household size and indicators for being married, being divorced, and the number of children under 5. Education controls include indicators for non-academic tertiary degree, vocational education, university entrance diploma, compulsory education, and university degree. MES refers to the [Cragg and Donald \(1993\)](#) minimum eigenvalue statistic. Data source: Swiss Labor Force Survey (SLFS) combined with data specifically collected in connection with the voucher experiment, which was conducted in 2006.

* p<0.10.

** p<0.05.

*** p<0.01.

Table 7
The effect of adult education on subsequent adult education.

	OLS				IV	
	(1)	(2)	(3)	(4)	1st stage (5)	2nd stage (6)
Self-financed adult education in 2006	.311*** (.011)	.306*** (.011)	.282*** (.011)	.226*** (.011)		.106 (.099)
Voucher in 2006					.127*** (.011)	
Firm training in 2006	.341*** (.012)	.336*** (.012)	.315*** (.012)	.246*** (.012)	–.058*** (.010)	.239*** (.013)
Adult education in 2005				.232*** (.009)	.172*** (.008)	.252*** (.019)
Woman	.002 (.009)	.003 (.009)	.027*** (.009)	.023** (.009)	.106*** (.008)	.036*** (.014)
Swiss	.100*** (.013)	.014 (.014)	.050*** (.015)	.040*** (.014)	.035*** (.012)	.044*** (.015)
Age	.019*** (.004)	.023*** (.004)	.021*** (.004)	.016*** (.004)	.002 (.003)	.016*** (.004)
Age ² /10 ³	–.252*** (.041)	–.289*** (.042)	–.262*** (.042)	–.200*** (.041)	–.037 (.036)	–.205*** (.041)
Region	Yes	Yes	Yes	Yes	Yes	Yes
Language	No	Yes	Yes	Yes	Yes	Yes
Family	No	Yes	Yes	Yes	Yes	Yes
Education	No	No	Yes	Yes	Yes	Yes
Observations	10,515	10,515	10,515	10,515	10,515	10,515
R-squared	.169	.174	.191	.235	.111	.227
MES	–	–	–	–	134.58	–

Note: Dependent variable: adult education in 2007 (in column 5: self-financed adult education in 2006). Standard errors in parentheses. Regional controls include 40 dummies for region of residence. Family controls include household size and indicators for being married, being divorced, and the number of children under 5. Education controls include indicators for non-academic tertiary degree, vocational education, university entrance diploma, compulsory education, and university degree. MES refers to the [Cragg and Donald \(1993\)](#) minimum eigenvalue statistic. Data source: Swiss Labor Force Survey (SLFS) combined with data specifically collected in connection with the voucher experiment, which was conducted in 2006.

* p<0.10.

** p<0.05.

*** p<0.01.

show that the coefficient on self-financed adult education in 2006 is further reduced to 0.23, but remains large and statistically highly significant.

To test whether voucher-induced adult education indeed leads to subsequent private human capital investments, we again instrument self-financed adult education by voucher receipt. The estimated IV effect of adult education in 2006 on adult education in 2007 is 0.106 and statistically insignificant (column 6 of Table 7). The standard errors are, however, again large, so that the effect might still be economically significant. The point estimate suggests that the probability of subsequent private investment in adult education is raised by 11 percentage points through voucher-induced adult education in 2006. This is, however, well below the OLS estimates in columns 1 to 4. Despite the somewhat low precision of the IV estimate, we can statistically reject the large effects found in the cross-sectional OLS estimations.

The positive point estimate for voucher-induced education in 2006 in predicting participation in adult education in 2007 also mitigates concerns that our labor-market estimates are overestimated due to voucher-induced intertemporal shifts in course participation. If there were any shifts from participation planned anyways for 2007 into 2006 due to the voucher option, this was over-compensated by any positive effect of education participation on subsequent education participation.

Summing up, the point estimates of the IV models raise considerable doubt that voucher-induced participation in adult education courses has substantial effects on labor market success. OLS estimates with different sets of controls vary widely, indicating that omitted variables problems are important—but possibly also reflecting that there are heterogeneous effects of adult education on labor market success in the population and selective participation in adult education. The IV estimates suggest that there may be small positive effects of voucher-induced adult education on subsequent private investments, but these effects are not of the magnitude suggested by cross-sectional OLS estimates.

6. The potential of untargeted vs. targeted programs: evidence on effect heterogeneity

Results on the average impact of the adult education voucher program cast strong doubt on the effectiveness of untargeted voucher programs. Our experimental setup, however, allows us to investigate whether the voucher program affects different sub-populations differently, in terms of both labor market returns and voucher take-up. This provides some indication on whether targeted voucher programs have the potential to be more effective.

6.1. Who benefits from the voucher treatment?

We start our investigation of heterogeneous effects of the voucher program by looking at labor market returns for different sub-populations (Table 8). While limited statistical power in the reduced samples limits strong conclusions, there is a clear indication that if there is any effect of the voucher program on earnings, it is likely to be restricted to the group of individuals whose highest educational degree is one of vocational training (column 2). While the point estimate is negative for all other groups, the estimated returns for individuals with vocational training is positive and reaches marginal significance at the 10% level.

Unfortunately, given the statistical imprecision of IV estimates already in the full sample, IV estimates for sub-populations are basically uninformative about effect heterogeneity (not shown). However, under the assumption that any remaining omitted variable bias in the OLS models does not differ across sub-populations, the more precise OLS models are informative about the relative size of the

Table 8

The impact of the voucher program by sub-groups.

	Self-financed adult education in 2006 (1)	Earnings in 2007 (2)	Employed in 2007 (3)	Adult education in 2007 (4)
Compulsory education	.0692*** (.025)	−.0188 (.027)	.00286 (.028)	.0208 (.033)
Vocational training	.114*** (.015)	.0159* (.009)	−.000493 (.012)	.00441 (.019)
University entrance diploma	.162*** (.040)	−.0213 (.032)	−.0182 (.031)	.00336 (.043)
Non-academic tertiary	.172*** (.028)	−.0248 (.017)	−.00119 (.014)	.0107 (.030)
University	.156*** (.035)	−.0135 (.028)	.000660 (.021)	−.00423 (.040)
Men	.0973*** (.015)	−.00592 (.009)	.00373 (.010)	.00959 (.019)
Women	.152*** (.016)	.00789 (.011)	−.00673 (.012)	.0141 (.017)

Note: ITT estimates of having received a voucher in 2006 for subgroups of the population defined by the characteristic reported in the first column. The first row indicates the dependent variable. Standard errors in parentheses. The second column reports first-stage results by subgroups based on the first-stage specification used in Table 7. Otherwise, specifications correspond to the specifications presented in Table 3. Data source: Swiss Labor Force Survey (SLFS) combined with data specifically collected in connection with the voucher experiment, which was conducted in 2006.

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

effects of adult education for different sub-populations. Therefore, Table 9 reports OLS estimates based on Eq. (3), which includes pre-treatment controls and the lagged outcome variables, for different sub-populations.

The OLS estimates of returns to adult education in terms of future earnings show a decreasing pattern by highest degree of education completed. They are significantly positive for the lowest two educational categories, but the point estimates turn negative for individuals with tertiary degrees. The result is in line with theoretical models assuming decreasing private returns to human capital investments. It

Table 9

The effect of adult education by sub-groups: OLS estimates with lagged dependent variables.

	Earnings in 2007 (1)	Employed in 2007 (2)	Adult education in 2007 (3)
Compulsory education	.0498* (.028)	.0300 (.029)	.192*** (.034)
Vocational training	.0182** (.009)	.00633 (.011)	.269*** (.017)
University entrance diploma	.0173 (.032)	.0342 (.027)	.155*** (.037)
Non-academic tertiary	−.00802 (.015)	.0133 (.012)	.195*** (.026)
University	−.0102 (.020)	.0221 (.015)	.199*** (.029)
Men	.00524 (.010)	.00994 (.010)	.210*** (.019)
Women	.0124 (.009)	.0150 (.010)	.235*** (.014)

Note: Results report estimates for the effect of participating in self-financed adult education in 2006. The OLS estimates are reported for subgroups of the population defined by the characteristic reported in the first column. The first row indicates the dependent variable. Standard errors in parentheses. All estimates are based on the specifications reported in the fourth columns of Tables 5–7. Data source: Swiss Labor Force Survey (SLFS) combined with data specifically collected in connection with the voucher experiment, which was conducted in 2006.

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

may also reflect that individuals with lower schooling have higher discount rates, so that they are only willing to take an education course if returns are very high (see Card, 1995). The OLS point estimates also suggest that the effects of adult education on future earnings and employment are stronger for women than for men.

In sum, the general pattern of ITT and OLS estimates on heterogeneous effects is quite consistent. The sub-sample analysis suggests that significant effect heterogeneity is hidden behind the average effects analyzed in the previous section. In particular, the effect of self-financed adult education on earnings may be largest for those with relatively low – in particular, vocational – education, and lowest for highly educated individuals.

6.2. Who takes up the vouchers? Characterizing the complier population

Not only the labor market returns, but also the propensity to use an offered voucher in the first place may differ by sub-population. From a policy perspective, it is important to understand which group of individuals will change their participation decision in response to the voucher. As a first indication on this question, the first column of Table 8 reveals that the effect of voucher receipt on participation in adult education indeed varies significantly by sub-group. Voucher take-up clearly increases with the level of educational attainment: the effect of voucher receipt on adult education participation is lowest at 6.9% for individuals with only compulsory schooling and highest at 16–17% for individuals with a university entrance diploma, non-academic or academic tertiary education. Pick-up rates are also significantly higher for women than for men.

We can gain further insight in take-up heterogeneity by characterizing the sub-population that complies with the voucher assignment based on several background characteristics. In the LATE framework, the individuals who are induced to change participation status by the voucher are the compliers. Naturally, because there are always-takers, compliers cannot be individually identified. However, it is possible to describe the distribution of characteristics of the compliers. This characterization is important for policy makers as it shows who will be ultimately affected by voucher programs. This characterization might differ from the descriptive statistics presented in Section 3, as the group averages presented there refer to the entire treatment group, which includes always-takers and never-takers. The latter populations are, however, not of direct interest from a policy perspective.

To characterize compliers based on the subset of Bernoulli-distributed characteristics, we can make use of the variation in the first-stage estimates across covariate groups, as shown in Angrist and Pischke (2009). Let x_{1i} be a background characteristic of interest, for example, a dummy indicating university graduates. The relative likelihood of a voucher complier to be a university graduate is given by the following calculation:

$$\begin{aligned} \frac{P[x_{1i} = 1 | D_{1i} > D_{0i}]}{P[x_{1i} = 1]} &= \frac{P[D_{1i} > D_{0i} | x_{1i} = 1]}{P[D_{1i} > D_{0i}]} \\ &= \frac{E[D_i | z_i = 1, x_{1i} = 1] - E[D_i | z_i = 0, x_{1i} = 1]}{E[D_i | z_i = 1] - E[D_i | z_i = 0]} \end{aligned}$$

where $z_i = 1$ indicates voucher receipt, D_{1i} denotes participation in adult education in case of voucher receipt, and D_{0i} denotes the counterfactual outcome of participation in adult education in case individual i does not receive a voucher. That is, the relative likelihood that a complier has $x_{1i} = 1$ is given by the ratio of the first stage for individuals with $x_{1i} = 1$ to the overall first stage.

Average characteristics for compliers are reported in Table 10. The final column shows the relative likelihood that compliers have the characteristic indicated in each row. It becomes apparent that individuals with higher educational degrees are also more likely to participate in adult education in response to receiving a voucher. Moreover,

Table 10
Characterizing compliers.

Variable	$P[x_{1i} = 1]$	$P[x_{1i} = 1 D_{1i} > D_{0i}]$	$P[x_{1i} = 1 D_{1i} > D_{0i}] / P[x_{1i} = 1]$
	(1)	(2)	(3)
Woman	0.56	0.66	1.19
Compulsory education	0.14	0.07	0.54
Vocational education	0.47	0.42	0.88
University entrance diploma	0.08	0.10	1.23
Non-academic tertiary	0.17	0.22	1.34
University	0.14	0.17	1.23
Age 20 to 29	0.12	0.07	0.56
Age 30 to 39	0.26	0.19	0.75
Age 40 to 49	0.29	0.38	1.31
Age 50 to 60	0.28	0.31	1.12
Employed in 2005	0.82	0.79	0.96
Adult education in 2005	0.44	0.47	1.07

Note: The table reports an analysis of complier characteristics. The ratio in the last column shows the relative likelihood compliers have the characteristic indicated in each row. Data source: Swiss Labor Force Survey (SLFS) combined with data specifically collected in connection with the voucher experiment, which was conducted in 2006.

women are more likely to comply with the voucher assignment than men, and compliance is higher in older age groups.

These findings – together with the evidence on heterogeneous benefits – might explain the low estimates of average labor market returns of voucher-induced adult education, as it points to a highly selective use of the voucher. On the one hand, individuals with higher levels of education are more likely to be induced to take up the voucher, but they appear to have below-average returns to adult education. Highly educated individuals are typically well informed about the returns to education and potentially have easier access to firm-financed adult education. Thus, highly educated compliers with the voucher assignment might be a selected sub-population of all individuals with tertiary education that face low labor market returns to adult education. Moreover, if private returns to human capital investments are decreasing in the level of human capital, individuals with tertiary education might more generally face low labor market returns to adult education. But highly educated individuals might draw direct utility from adult education, which could explain their high pick-up rates. On the other hand, individuals with less schooling are also less likely to be induced to participate in adult education when receiving a voucher. But individuals with lower levels of educational attainment, in particular with vocational training, appear to gain the most from voucher-induced adult education.

In sum, the evidence on heterogeneity in the returns to and take-up of the vouchers suggests that programs that are targeted at – and effective in inducing – less-educated individuals may have a greater potential to be effective than the untargeted voucher program studied in this paper. However, it is unclear what might help to induce less-educated individuals to in fact use the voucher to participate in adult education courses. With the untargeted intervention, the main group of individuals who are induced to increasingly participate in higher education are individuals with tertiary education, but the labor market returns to additional participation in this group appear particularly small.

7. Crowd-out of firm-financed training

Apart from the limited extent to which the untargeted voucher program may reach particular sub-populations of policy interest, another possible concern with voucher programs from a public policy perspective is that the publicly-funded voucher-induced course participation may crowd out firm-financed adult education. While the descriptive pattern in Table 2 suggested an insignificant, albeit negative difference in the participation in firm-financed adult education between the treatment and control group of the voucher experiment,

Table 11
Substitution of firm-financed training.

	ITT	OLS				IV	
	(1)	(2)	(3)	(4)	(5)	1st stage (6)	2nd stage (7)
Self-financed adult education in 2006		.012 (.009)	.008 (.009)	–.010 (.010)	–.056*** (.009)		–.185** (.083)
Voucher in 2006	–.024** (.011)					.129*** (.011)	
Adult education in 2005	.187*** (.008)				.196*** (.008)	.161*** (.008)	.217*** (.015)
Woman	–.054*** (.007)	–.068*** (.008)	–.067*** (.008)	–.047*** (.008)	–.048*** (.008)	.109*** (.008)	–.034*** (.012)
Swiss	.017 (.012)	.073*** (.011)	.044*** (.012)	.027** (.012)	.019 (.012)	.034*** (.012)	.024* (.012)
Age	.012*** (.003)	.017*** (.003)	.020*** (.003)	.017*** (.003)	.012*** (.003)	.001 (.003)	.012*** (.003)
Age ² /10 ³	–.155*** (.035)	–.210*** (.034)	–.251*** (.036)	–.203*** (.035)	–.156*** (.035)	–.028 (.036)	–.160*** (.035)
Region	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Language	Yes	No	Yes	Yes	Yes	Yes	Yes
Family	Yes	No	Yes	Yes	Yes	Yes	Yes
Education	Yes	No	No	Yes	Yes	Yes	Yes
Observations	10,520	10,521	10,521	10,521	10,520	10,520	10,520
R-squared	.100	.026	.032	.046	.102	.108	.086
MES	–	–	–	–	–	138.45	–

Note: Dependent variable: Firm-financed training dummy in 2006 (in column 5: self-financed adult education in 2006). Standard errors in parentheses. Regional controls include 40 dummies for region of residence. Family controls include household size and indicators for being married, being divorced, and the number of children under 5. Education controls include indicators for non-academic tertiary degree, vocational education, university entrance diploma, compulsory education, and university degree. MES refers to the [Cragg and Donald \(1993\)](#) minimum eigenvalue statistic. Data source: Swiss Labor Force Survey (SLFS) combined with data specifically collected in connection with the voucher experiment, which was conducted in 2006.

* $p < 0.10$.

** $p < 0.05$.

*** $p < 0.01$.

more fully controlled estimation may yield additional insight in this question. In fact, the first column of [Table 11](#) suggests that once our control variables are held constant, the negative effect of voucher receipt on contemporaneous participation in firm-financed training courses in fact becomes statistically significant. The ITT point estimate suggests that voucher receipt reduces participation in firm-financed training by 2.4 percentage points.

Similarly, while the OLS estimates of the effect of participation in self-financed adult education in 2006 on participation in firm-financed adult education in 2006 in models without extensive controls are statistically insignificant, they in fact become significantly negative once the lagged dependent variable (participation in adult education in 2005) is controlled for (column 5). Similarly, the IV estimate of the effect of voucher-induced self-financed education on firm-financed education over the same period is significantly negative.

These results suggest that there is, in fact, significant crowd-out of firm-financed training by the voucher program. Part of the voucher-induced participation in adult education simply seems to substitute for participation that would otherwise have been financed by employers. From a public policy viewpoint, this reduces the potential social benefits of a voucher program. In particular, if the firm-based training that is crowded out is replaced by less focused – and potentially less productive – forms of adult education, the effect of the voucher program on labor market outcomes may indeed be negative overall. Given that firm-financed training is the primary source of human capital accumulation for adult workers (e.g., [de Grip and Sauermann, forthcoming](#)), a reduction of this form of training may tip the policy evaluation of a public subsidy for self-financed adult education to the negative.

8. Conclusion

In this paper, we evaluate the effects of a large-scale field experiment that issued vouchers for adult education to a randomly devised treatment group in a large sample representative of the Swiss adult

population. Apart from the specific literature on public-sector training programs targeted at the unemployed as part of active labor market policies, this is the first experimental evidence on the returns to adult education. By contrast, the extensive existing literature on returns to employer-provided adult education and training so far suffers from a lack of randomization. Our experiment sheds light on governments' ability to improve labor market outcomes through untargeted voucher programs aimed at expanding adult education, and it provides some indication of the possibly higher potential of targeted programs.

We find no significant average labor market effects of the voucher program. In particular, while the voucher program indeed increased participation in self-financed adult education courses, the take-up rate was so small as to make any substantial aggregate labor market effects unlikely. In fact, the point estimates of the short-term effects of the voucher program effective in the first half of 2006 on earnings and employment in 2007 are not only statistically indistinguishable from zero, but also negative. Using an IV approach that instruments participation in adult education by voucher receipt, we find no significant average effects of voucher-induced participation in adult education on earnings and employment. Unfortunately, the IV estimates lack precision, so we cannot reject the existence of small average effects of adult education. However, all IV point estimates are close to zero and well below corresponding OLS estimates of the effects of adult education.

Investigating voucher effects on subsequent adult education, we reject the possibility that the large persistence in adult education implied by cross-sectional OLS estimation is causal. Small positive effects of voucher-induced adult education on subsequent human capital investments cannot be ruled out and are in fact suggested by the IV point estimates.

However, it seems that a noteworthy part of the voucher-induced participation in self-financed education comes at the detriment of otherwise firm-financed training courses. Close to one in five individuals who use the voucher to pay for a course appear to not participate

in a firm-financed course instead. To the extent that firm-financed training may well be the most focused and productive source of accumulation of human capital in the adult population, this raises concerns that public voucher programs may in fact have negative aggregate effects on labor market outcomes. Furthermore, evidence of substantial deadweight losses due to always-takers adds to the cost side of an evaluation of public voucher programs.

In terms of the effect of adult education on labor market success, estimated OLS effects decrease significantly when adding covariates and, in particular, lagged dependent variables. This indicates important omitted variable problems in standard analyses. Furthermore, participation in adult education is highly selective and returns may be highly heterogeneous, which may also be a cause for the relatively low statistical precision of our IV estimates of the average effect. A sub-group analysis reveals that especially individuals with low levels of education might benefit from participating in adult education. But a characterization of the complier population shows that predominantly individuals with higher levels of education take up the voucher to participate in adult education. There are several possible sources for this tension of the largest voucher take-up by those for whom returns are lowest. High-educated people may derive direct “consumption” utility from attending adult education courses; low-educated people may have particularly high discount rates; low-educated people may also face particular constraints for adult education or lack the correct information on possible returns; and finally, the tension might also reflect irrational behavior. The relative importance of these (or other) explanations is an important question for future research.

Our findings are relevant for guiding policies to promote adult education by voucher programs. The results cast strong doubt that voucher programs that are neither targeted at specific groups nor restricted to specific approved uses are effective in improving labor market outcomes. In particular, the potential of voucher programs to generate large positive average effects on labor market outcomes is questionable. In addition, the characterization of the group of compliers is helpful in a policy perspective. Only the complying individuals will change their participation decision in adult education in response to a voucher program. In the case of an untargeted voucher program, the voucher will mainly affect the behavior of highly educated individuals. This population might, however, not be the primary target for government interventions in the adult education market. Our results suggest that individuals with lower levels of educational attainment might benefit more from voucher-induced adult education. A specifically designed and targeted voucher program might, therefore, potentially generate larger effects on subsequent labor market success.

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